A2 Report

1. EDA Insights

- **Dataset:** 272 discharge summaries (204 train/68 test), 74,808 entity annotations, 6,624 unique SNOMED CT concepts.
- **Note length:** 2,374–24,858 chars (median 9,663). Median 2,804 tokens → all notes exceed a single 512-token context; chunking is required.
- **Entity density:** ~275 entities per note (median 267). Very dense annotation.
- **Entity length:** median 8 chars, mean 9.46; 75th percentile 12 chars. Short spans dominate.
- **Label distribution:** ~70.7% O, 10.4% B, 18.9% I → standard cross-entropy loss is appropriate; no class weighting needed.
- **Top concepts:** dominated by frequent findings ("No abnormality detected" 905×, "Blood pressure" 674×), long-tail distribution.

2. Experimental Setup

Approach A — Encoder NER + Linker (SNOMED)

- NER: `emilyalsentzer/Bio_ClinicalBERT`, BIO tagging, 512-token windows with 128-token stride, fp16, DDP across 4 GPUs.
- Retrieval: SapBERT over SNOMED CT, concept-level max-pool by code.
- Linker: cross-encoder fine-tuned on `(entity, concept)` pairs from SapBERT candidates.

Approach B — Decoder LLM (BioMistral-7B)

- **Few-shot:** base BioMistral-7B, prompt with 2 examples and greedy decoding.
- **QLoRA:** 4-bit NF4 base + LoRA (r=16, α=32) on all linear projections, 589 training chunks, 3 epochs, `paged_adamw_8bit`.

Approach C — UMLS-based Clinical Concept Mapping

- UMLS preprocessing: English, non-suppressed, filtered SABs (SNOMEDCT_US, MSH, RXNORM, LNC, HPO, OMIM, DRUGBANK), deduplicated to 3.03M `(CUI, cleaned STR)` rows.
- Retrieval: SapBERT (fp16, ~4.6 GB index) over full UMLS, concept-level max-pool by CUI.
- Linker: cross-encoder retrained on UMLS candidate pairs.

- Context handling: rule-based NegEx-style assertion classifier (negation / historical / hypothetical) applied to each extracted span.

3. Evaluation Metrics

- **Entity Recognition:** exact-span Precision, Recall, F1.
- **Concept Linking:** Accuracy. Reported against gold SNOMED CT IDs for A, B1, B2 and against UMLS CUIs for C.
- **End-to-End:** Precision, Recall, F1.

4. Results

Entity Recognition

Approach	Precision	Recall	F1
A — Encoder NER (Bio_ClinicalBERT)	0.7663	0.8544	0.8079
B1 — BioMistral Few-Shot	0.3544	0.0048	0.0095
B2 — BioMistral QLoRA	0.6879	0.0344	0.0656
C — Full UMLS Pipeline	0.7771	0.8540	0.8137

Concept Linking

Approach	Accuracy (SNOMED ID)	Accuracy (UMLS CUI)
A — Encoder + Linker	0.5193	—
B1 — BioMistral Few-Shot	0.0893	—
B2 — BioMistral QLoRA	0.7312	—
C — Full UMLS Pipeline	0.5660	0.6096

End-to-End Pipeline

Approach	Precision	Recall	F1
A — Encoder + Linker	0.3979	0.4437	0.4196
B1 — BioMistral Few-Shot	0.0316	0.0004	0.0008
B2 — BioMistral QLoRA	0.5009	0.0252	0.0479
C — Full UMLS Pipeline	0.4738	0.5214	0.4965

5. Comparative Analysis

- **Encoder vs decoder:** the encoder pipeline worked a lot better. BioMistral few-shot essentially fails: the base model rarely emits valid JSON with correct SNOMED IDs at this prompt length (F1 0.009). QLoRA lifts linking accuracy to 0.73 and end-to-end precision to 0.50, but recall collapses to 0.03 as the model learns the output *format* but misses most entities in heavy chunks.
- **SNOMED-only (A) vs full UMLS (C):** extending retrieval from SNOMED-only to full UMLS with CUI as the target gives **+9.0 linking accuracy** and **+7.7 end to end F1**. CUI accuracy (0.610) exceeds SNOMED ID accuracy (0.566) probably due to some concepts being correctly identified through non-SNOMED sources even when the SNOMED mapping disagrees with gold.
- **Task 3 pipeline dominance:** Approach C works better at every stage except precision on B2's near-empty predictions (which is a degenerate case caused by extremely low recall).

6. Error Analysis & Limitations

Encoder / UMLS pipelines:

- **Retrieval ceiling:** SapBERT top-20 recall on gold text **≈ 0.72** — this caps end-to-end F1 regardless of linker quality. Main retrieval failures: abbreviations (`"ra"` , `"nad"` , `"vs"`) and newline-containing spans.
- **Annotation noise:** gold includes semantically degenerate spans (`"vs"` , `"improved"`) that are intrinsically unretrievable.
- **Span boundary errors:** chunked NER occasionally splits entities across chunk edges.

Decoder pipelines:

- **Few-shot failure modes:** prose instead of JSON, hallucinated SNOMED IDs, truncated JSON.
- **QLoRA failure modes:** low entity recall — 589 training examples are insufficient to teach a 7B model to extract ~275 entities per note; instruction-tuned base is unaccustomed to this format.
- **Compute ceiling:** 4-bit QLoRA on 11 GB constrains context length and batch size.

Context handling (negation/temporality) is rule-based and *tags* rather than filters negated mentions, so linking metrics include them.
UMLS index restricted to 7 SABs to fit on 11 GB GPUs.

7. Conclusion

The encoder pipeline with SapBERT retrieval and a cross-encoder linker is the strongest approach. Extending candidate generation from SNOMED-only to full UMLS and switching the target from SNOMED codes to UMLS CUIs raises end-to-end F1 from 0.42 to 0.50. Decoder-based LLMs (BioMistral-7B) are viable only as rerankers on top of retrieved candidates; without retrieval, they fail to produce valid structured output at this scale. The dominant remaining bottleneck is retrieval recall (~0.72), driven by abbreviation-heavy clinical text and annotation noise.